

STAT 7650: Computational Statistics

9. Advanced Topics in MCMC

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Outline

- 1 Adaptive MCMC
- 2 Reversible jump MCMC

Motivation

In a typical implementation of a Metropolis-Hastings algorithm,

- Target distribution $f(x)$
- Proposal distribution $g(x | x^{(t)})$
- Acceptance ratio R

$$R = \frac{f(x^*)g(x^{(t)} | x^*)}{f(x^{(t)})g(x^* | x^{(t)})}$$

- Tune $g(x | x^{(t)})$ to make sure the acceptance rate in $(0.25, 0.50)$
 - $g(x | x^{(t)})$ usually depends on some tuning parameters
 - Try pilot runs with different choices of tuning parameters
 - Becomes more challenging for high-dimensional problems

Adaptive MCMC

Adaptive MCMC algorithms allow for automatic tuning of the proposal distribution as iterations progress.

- **Diminishing adaption:** As $t \rightarrow \infty$, the parameters in g will depend less and less on earlier states of the chain.
 - Modify the parameters in g by smaller amount as t increases.
 - Make the adaptations less frequently as t increases.
- **Bounded convergence:** $M^{(t)}(\epsilon)$ is bounded in probability, $\epsilon > 0$.
 - $D^{(t)}$: the total variation distance between the stationary distribution of the transition kernel of the AMCMC algorithm at time t and the target stationary distribution.
 - $M^{(t)}(\epsilon)$: the smallest t such that $D^{(t)} < \epsilon$.

Adaptive Random Walk Metropolis-within-Gibbs

Consider a Gibbs sampler,

- The univariate conditional density for the first element of X is not available in closed form.
- Use a random walk Metropolis algorithm.

$$x_1^* = x_1^{(t)} + \epsilon, \quad \epsilon \sim N(0, \sigma_b^2)$$

- The univariate conditional densities for the remaining elements of X permit standard Gibbs updates.

The goal of the AMCMC algorithm is to tune the variance of the proposal distribution so that the acceptance rate is optimal.

- The adaption step is performed at T_b , where $b = 0, 1, \dots$
 - For example, $T_1 = 50, T_2 = 100, T_3 = 150, \dots$
 - Change the proposal distribution variance σ_b^2 at step T_b

Algorithm

- Initialization: set $t = 0$ and batch index $b = 0$
 - Select starting values $X^{(0)} = x^{(0)}$ and let $\sigma_0^2 = 1$
 - Select a batching schedule $\{T_b\}$ for $b = 0, 1, 2, \dots$
- Metropolis-within-Gibbs update: update $X_1^{(t)}$
 - Generate X_1^* by drawing $\epsilon \sim N(0, \sigma_b^2)$ and set $X_1^* = x_1^{(t)} + \epsilon$
 - Compute the MH ratio $R(x_1^{(t)}, X_1^*) = f(x_1^*)/f(x_1^{(t)})$
 - Sample a value for $X_1^{(t+1)}$ according to the following

$$X_1^{(t+1)} = \begin{cases} X_1^*, & \text{with probability } \min\{R(x_1^{(t)}, X_1^*), 1\} \\ x_1^{(t)}, & \text{otherwise} \end{cases}$$

- Gibbs update: update $X_i^{(t)}$ for $i = 2, \dots, p$ in turn
- Adaption step: when $t = T_{b+1}$, update σ_b^2 , and increment b

$$\log \sigma_{b+1} = \log \sigma_b \pm \delta(b+1)$$

Comments

- Add the adaption factor $\delta(b+1)$ when the acceptance rate is smaller than 0.44 for the iterations in the previous batch and subtract it otherwise.
- A common choice is $\delta(b+1) = \min(0.01, 1/\sqrt{T_b})$
- The adaptive algorithm can be generalized to other random walk distributions.
- It is particularly useful when there are many parameters, each with its own variance to be tuned.

Example: Whale Population Dynamics

The Pella-Tomlinson model is a population dynamics model, which describes changes in animal abundance over time.

$$N_{y+1} = N_y - C_y + rN_y \left(1 - \left(\frac{N_y}{K} \right)^2 \right)$$

where

- N_y : animal abundance in year y
- C_y : catch (human-based removals) in year y
- r : the intrinsic growth rate
- K is the carrying capacity, which is the number of animals that can be sustained in equilibrium with the amount of resources available within the limited range inhabited by the population.

Data and Prior Distributions

- The dataset contains C_y for 101 year
- Observed estimates of N_y at $y \in \{14, 21, 63, 91, 93, 100\}$

$$\log \hat{N}_y \sim N(\log(N_y), \hat{\sigma}_y^2)$$

where $\hat{\sigma}_y^2 = \log(1 + \psi_y^2)$ and $\psi_y = \psi$ does not depend on y .

- Assume $N_0 = K$, the population is in equilibrium at $y = 0$
- The goal is to estimate three parameters K , r , and ψ

Consider the prior, which are based on historical researches

$$K \sim \text{unif}(7,000, 100,000), \quad r \sim \text{unif}(0.001, 0.1), \quad \psi/2 \sim \text{beta}(2, 10)$$

Proposal Distributions

We use a hybrid Gibbs approach. Let Gibbs cycles be indexed by t . The proposals for each parameters at cycle $t + 1$ are

$$K^* = K^{(t)} + \epsilon_K^{(t+1)}, \quad r^* = r^{(t)} + \epsilon_r^{(t+1)}, \quad \psi^* = \psi^{(t)} + \epsilon_\psi^{(t+1)}$$

where the proposal distributions are

$$g_{\epsilon_K}(\epsilon_K^{(t+1)} \mid K^{(t)}) \propto \phi(\epsilon_K^{(t+1)}; 0, 200) I(\epsilon_K^{(t+1)} \in S_K)$$

$$g_{\epsilon_r}(\epsilon_r^{(t+1)} \mid r^{(t)}) \propto I(\epsilon_r^{(t+1)} \in S_r)$$

$$g_{\epsilon_\psi}(\epsilon_\psi^{(t+1)} \mid \psi^{(t)}) \propto \phi(\epsilon_\psi^{(t+1)}; 0, 0.1) I(\epsilon_\psi^{(t+1)} \in S_\psi)$$

where the supports are

$$S_K = \{\epsilon_K^{(t+1)} : 7000 \leq K^{(t)} + \epsilon_K^{(t+1)} \leq 100000\}$$

$$S_r = \{\epsilon_r^{(t+1)} : 0.001 \leq r^{(t)} + \epsilon_r^{(t+1)} \leq 0.1, -0.03 \leq \epsilon_r^{(t+1)} \leq 0.03\}$$

$$S_\psi = \{\epsilon_\psi^{(t+1)} : 0 < \psi^{(t)} + \epsilon_\psi^{(t+1)} \leq 2\}$$

Standard Algorithms and Results

At the iteration t of a standard Metropolis-Hastings algorithm,

- Sample from the increment distribution
- Generate $K^{(t+1)}$ as follows:
 - Propose $K^* = K^{(t)} + \epsilon_K^{(t+1)}$.
 - Calculate R_K .
 - Set $K^{(t+1)} = K^*$ with probability $\min\{1, R_K\}$. Otherwise $K^{(t+1)} = K^{(t)}$.
- Similar steps for $r^{(t+1)}$ and $\psi^{(t+1)}$.

Generate a chain of length 45,000 with a burn-in of 10,000,

- The proposal distributions are not symmetric due to truncation. It is not a random walk. Need g when calculating R_K .
- The mixing properties of this chain are poor.
- The acceptance rates are 81%, 27%, and 68%.

Adaptive MCMC

Redefine the proposal distributions as

$$\begin{aligned}
 g_{\epsilon_K}^{(t+1)}(\epsilon_K^{(t+1)} \mid K^{(t)}) &\propto \phi(\epsilon_K^{(t+1)}; 0, 200\delta_K^{(t+1)})I(\epsilon_K^{(t+1)} \in S_K) \\
 g_{\epsilon_r}^{(t+1)}(\epsilon_r^{(t+1)} \mid r^{(t)}) &\propto I(\epsilon_r^{(t+1)} \in S_r^{(t+1)}) \\
 g_{\epsilon_\psi}^{(t+1)}(\epsilon_\psi^{(t+1)} \mid \psi^{(t)}) &\propto \phi(\epsilon_\psi^{(t+1)}; 0, 0.1\delta_\psi^{(t+1)})I(\epsilon_\psi^{(t+1)} \in S_\psi)
 \end{aligned}$$

where the supports are

$$S_r^{(t+1)} = \{\epsilon_r^{(t+1)} : 0.001 \leq r^{(t+1)} \leq 0.1, |\epsilon_r^{(t+1)}| \leq 0.03\delta_r^{(t+1)}\}$$

and $\delta_K^{(t+1)}$, $\delta_r^{(t+1)}$, and $\delta_\psi^{(t+1)}$ are adaptation factors that vary as t increases.

We adjust $\delta_K^{(t+1)}$, $\delta_r^{(t+1)}$, and $\delta_\psi^{(t+1)}$ every 1500 iteration.

$$\log \delta_K^{(t+1)} = \log \delta_K^{(t)} + \frac{u_K^{(t+1)}}{(t+1)^{1/3}}$$

$$\log \delta_r^{(t+1)} = \log \delta_r^{(t)} + \frac{u_r^{(t+1)}}{(t+1)^{1/3}}$$

$$\log \delta_\psi^{(t+1)} = \log \delta_\psi^{(t)} + \frac{u_\psi^{(t+1)}}{(t+1)^{1/3}}$$

The goal is to achieve acceptance rates near 44%. At a specific step t_a , compute the acceptance rate during the previous 1500 iterations,

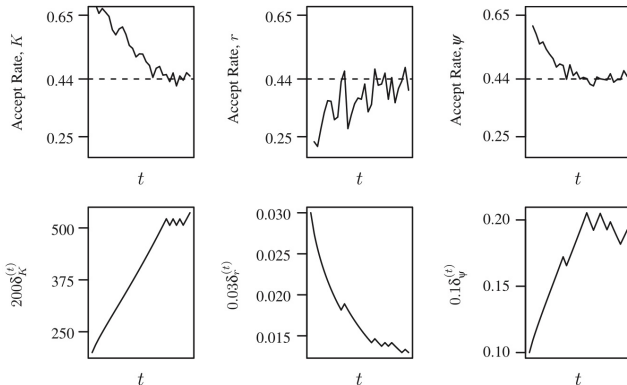
$$u_\theta^{(t_a+1)} = \begin{cases} -1, & \text{acceptance rate} < 44\% \\ 1, & \text{otherwise} \end{cases}$$

Results

If $t \in \{1500, 3000, \dots, 42000\}$, then for each parameter,

- Calculate the acceptance rate in the most recent 1500 iterations
- Determine $u^{(t+1)}$, update $\delta^{(t+1)}$, and get $g_\epsilon^{(t+1)}$

For other t , $g_\epsilon^{(t+1)}$ remain unchanged from the previous iteration.



Adaptive Metropolis Algorithm

The goal of the **adaptive Metropolis algorithm** is to estimate the variance of the proposal distribution during the algorithm.

- Random walk with a normal proposal distribution $N(X^{(t)}, \lambda \Sigma^{(t)})$
- The goal is to adapt $\Sigma^{(t)}$ during the algorithm
- When the target distribution is normal with covariance Σ_π , the optimal covariance of the proposal distribution is $(2.38^2/p)\Sigma_\pi$.
- Set $\lambda = (2.38^2/p)$ and estimate Σ_π based on previous iterations.

Algorithm

Initialization: set $t = 0$,

- Select $X^{(0)}$ at random, and set $\mu^{(0)} = 0$ and $\Sigma^{(0)} = I$.

Given $X^{(t)} = x^{(t)}$, $\mu^{(t)}$, and $\Sigma^{(t)}$, generate $X^{(t+1)}$ as follows.

- Sample $X^* \sim N(X^{(t)}, \lambda \Sigma^{(t)})$, where $\lambda = (2.38^2/p)$.
- Select the value $X^{(t+1)}$ according to

$$X^{(t+1)} = \begin{cases} X^*, & \text{with probability } \min\{R(x^{(t)}, X^*), 1\} \\ x^{(t)}, & \text{otherwise} \end{cases}$$

- Update the proposal distribution variance

$$\mu^{(t+1)} = \mu^{(t)} + \gamma^{(t+1)}(X^{(t+1)} - \mu^{(t)})$$

$$\Sigma^{(t+1)} = \Sigma^{(t)} + \gamma^{(t+1)}[(X^{(t+1)} - \mu^{(t)})(X^{(t+1)} - \mu^{(t)})^T - \Sigma^{(t)}]$$

where $\gamma^{(t+1)}$ is an adaption parameter with values chosen by the user. For example, $\gamma^{(t+1)} = 1/(t+1)$.

Comments

- To uphold the diminishing adaptation condition, it is required that $\lim_{t \rightarrow \infty} \gamma^{(t)} = 0$. The additional condition $\sum_{t=0}^{\infty} \gamma^{(t)} = \infty$ allows the sequence $\Sigma^{(t)}$ does not depend on its initial value.
- To further improve the algorithm, we can adapt λ ,

$$\log \lambda^{(t+1)} = \log \lambda^{(t)} + \gamma^{(t+1)}(R(x^{(t)}, X^*) - a)$$

where a is the target acceptance rate.

- For a high dimensional problem, each component may have its own scaling parameter $\lambda^{(t+1)}$ and is accepted/rejected separately. In this case, set a to a larger value, for example $a = 0.44$.
- It is possible to carry out the adaption in batches instead of every iteration. For example, only at $t = 50, 100, 150, \dots$ with fixed interval or $t = 50, 150, 300, 500, \dots$ with an increasing intervals.

Reversible Jump MCMC: Motivation

Consider a model selection problem,

- Observed data y
- Potential models $M_1, \dots, M_K$ with prior $f(m)$
- Parameter θ_m with prior $f(\theta_m | m)$ for the m th model
- The dimension p_m varies from different models

The random variable is $X = (M, \theta_M)$. The posterior density is

$$f(m, \theta_m | y) \propto f(y | m, \theta_m) f(\theta_m | m) f(m)$$

We need to construct a Markov chain to explore a space of varying dimensions.

Reversible Jump MCMC

The key to the **reversible jump MCMC** (RJMCMC) algorithm is the introduction of auxiliary random variables at times t and $t + 1$ with dimensions chosen so that the augmented variables at times t and $t + 1$ have equal dimensions.

For two models M_1 and M_2 with $p_2 > p_1$,

- Let U_1 be an independent random variable and $\theta_2 = q_{1,2}(\theta_1, U_1)$.
- The inverse transformation $(\theta_1, U_1) = q_{1,2}^{-1}(\theta_2) = q_{2,1}(\theta_2)$.

Hence, follow the steps below to propose $X^* = (M^*, \theta_M^*)$.

- propose a model M^*
- propose an augmenting variable U if necessary
- propose θ^* conditional on M^* and U

Algorithm

Let the current state be $x^{(t)} = (m^{(t)}, \theta_{m^{(t)}}^{(t)})$. The next iteration is

- Sample a candidate model $M^* \mid m^{(t)}$ from $g(\cdot \mid m^{(t)})$.
- Generate $U \mid (m^{(t)}, \theta_{m^{(t)}}^{(t)}, m^*)$ from $h(\cdot \mid m^{(t)}, \theta_{m^{(t)}}^{(t)}, m^*)$. Let

$$(\theta_{m^*}^*, U^*) = q_{t,*}(\theta_{m^{(t)}}^{(t)}, U),$$

where $q_{t,*}$ is an invertible mapping from $(\theta_{m^{(t)}}^{(t)}, U)$ to $(\theta_{M^*}^*, U^*)$ and $p_{m^{(t)}} + p_U = p_{m^*} + p_{U^*}$.

- Compute the Metropolis-Hastings ratio R
- Accept the move to the model M^* with probability $\min\{R, 1\}$. If the proposal is accepted, set $X^{(t+1)} = (M^*, \theta_{M^*}^*)$. Otherwise, reject the candidate draw and set $X^{(t+1)} = x^{(t)}$.

Comments

- The Metropolis-Hastings ratio is computed as

$$\frac{f(m^*, \theta_{m^*}^* \mid y) g(m^{(t)} \mid m^*) h(u^* \mid m^*, \theta_{m^*}^*, m^{(t)})}{f(m^{(t)}, \theta_{m^{(t)}}^{(t)} \mid y) g(m^* \mid m^{(t)}) h(u \mid m^{(t)}, \theta_{m^{(t)}}^{(t)}, m^*)} |J(t)|$$

where $J(t)$ is the Jacobian matrix.

$$J(t) = \frac{dq_{t,*}(\theta, u)}{d(\theta, u)} \Big|_{(\theta, u) = (\theta_{m^{(t)}}^{(t)}, U)}$$

Example: Jumping between Two Simple Models

Consider a problem with $K = 2$ possible models:

$$M_1: \theta_1 = \alpha. \quad M_2: \theta_2 = (\beta, \gamma).$$

When moving from M_1 to M_2 , the current state is $(1, \theta_1) = (1, \alpha)$

- Generate $U \sim h(u \mid 1, \theta_1, 2)$ from some density h .
- Let $\beta = \alpha - U$ and $\gamma = \alpha + U$, so

$$q_{1,2} = (\alpha - u, \alpha + u), \quad |dq_{1,2}(\alpha, u)/d(\alpha, u)| = 2.$$

- The Metropolis-Hastings ratio is

$$R_{1 \rightarrow 2} = \frac{f(2, \beta, \gamma \mid y) g(1 \mid 2) h(u^* \mid 2, \theta_2, 1)}{f(1, \alpha \mid y) g(2 \mid 1) h(u \mid 1, \alpha, 2)} \cdot 2$$

where $h(u^* \mid 2, \theta_2, 1) \equiv 1$ because $2 \rightarrow 1$ is deterministic.

Example: Continued

When moving from M_2 to M_1 , the current state is $(2, \theta_2) = (2, \beta, \gamma)$

- $(\alpha, u) = q_{2,1}(\beta, \gamma) = ((\beta + \gamma)/2, (\beta - \gamma)/2)$ and

$$|dq_{2,1}(\beta, \gamma)/d(\beta, \gamma)| = 1/2$$

- The Metropolis-Hastings ratio is

$$R_{2 \rightarrow 1} = \frac{f(1, \alpha \mid y) g(2 \mid 1) h(u \mid 1, \alpha, 2)}{f(2, \beta, \gamma \mid y) g(1 \mid 2) h(u^* \mid 2, \theta_2, 1)} \cdot \frac{1}{2}$$

where $h(u^* \mid 2, \theta_2, 1) \equiv 1$ because $2 \rightarrow 1$ is deterministic.

Note the two ratios $R_{2 \rightarrow 1}$ and $R_{1 \rightarrow 2}$ are reciprocals of each other.

RJMCMC for Variable Selection in Regression

Consider a variable selection problem in linear regression,

- p predictors and hence $K = 2^p$ possible models
- Model m_k includes predictors indexed by $\mathcal{J}_k \subseteq \{1, \dots, p\}$.
- The observation model is

$$Y = X_{m_k} \beta_{m_k} + \epsilon, \quad \epsilon \sim N(0, \sigma^2 I)$$

where X_{m_k} is the design matrix restricted to predictors in m_k .

- The prior distributions are

$$f(m_k) = 2^{-p}, \quad \beta_{m_k} \mid \sigma^2 \sim N(\alpha_{m_k}, \sigma^2 V_{m_k}), \quad v\lambda/\sigma^2 \sim \chi_v^2$$

- The goal is to sample from the posterior

$$f(m, \beta_m \mid y) \propto f(y \mid m, \beta_m) f(\beta_m \mid m) f(m)$$

Algorithm

- At each iteration, a new model M^* is proposed by adding or removing one predictor from the current model $m^{(t)}$ with equal probability. The proposal density is

$$g(m^* | m^{(t)}) = 1/p,$$

since there are p possible one-variable changes.

- Given $M^* = m^*$,
 - Generate $U | (m^{(t)}, \beta_{m^{(t)}}^{(t)}, m^*) \sim h(\cdot | m^{(t)}, \beta_{m^{(t)}}^{(t)}, m^*)$.
 - Let U be the next value for the parameter vector
 - Let h be the posterior for $\beta_m | (m, y)$, namely $f(\beta_m | m, y)$
 - Set $\beta_{m^*}^* = U$ and $U^* = \beta_{m^{(t)}}^{(t)}$
 - Hence $q_{t,*}(\beta_{m^{(t)}}^{(t)}, U) = (\beta_{m^*}^*, U^*)$ with $J = 1$.
 - $g(m^{(t)} | m^*) = g(m^* | m^{(t)}) = 1/p$.

- The MH ratio is

$$\frac{f(m^*, \beta_{m^*}^* | y) f(\beta_{m^{(t)}}^{(t)} | m^{(t)}, y)}{f(m^{(t)}, \beta_{m^{(t)}}^{(t)} | y) f(\beta_{m^*}^* | m^*, y)} = \frac{f(y | m^*) f(m^*)}{f(y | m^{(t)}) f(m^{(t)})}$$

where $f(y | m)$ is the marginal likelihood.

- With conjugate priors, we can integrate out β_m and σ^2 in closed form to obtain the marginal likelihood $f(y | m)$, which is a non-central t distribution. This bypasses the need to simulate β_{m^*} and σ^2 for the proposed model.

Example: Baseball Salaries

In the baseball salaries example,

- Always include the intercept
- $p = 27$ predictors and $2^{27} \approx 1.34 \times 10^8$ candidate models.
- Prior of σ^2 : set $v = 2.58$ and $\lambda = 0.28$
- Prior of β_m :
 - $\alpha_m = (\hat{\beta}_0, 0, \dots, 0)$ is a vector of length p_m , where $\hat{\beta}_0$ is the LSE of the intercept from the full model.
 - $V_m = \text{diag}(s_y^2, c^2/s_1^2, \dots, c^2/s_p^2)$, where s_y^2 is the sample variance of y , s_i^2 is the sample variance of the i th predictor, and $c = 2.58$.

Run the RJMCMC algorithm for 200,000 iterations.

Example: Posterior Model Probability

The RJMCMC produces samples $(m^{(1)}, \dots, m^{(T)})$ from the posterior $f(m \mid \mathbf{y})$. The **posterior model probability (PMP)** of model m_k is estimated by

$$\hat{f}(m_k \mid \mathbf{y}) = \frac{1}{T} \sum_{t=1}^T \mathbf{1}(m^{(t)} = m_k)$$

The five models with the highest estimated PMP are

Predictors							PMP
3	4	8	10	13	14	24	
•		•	•	•	•		0.22
	•	•	•	•	•		0.08
	•	•		•	•		0.05
•	•	•	•	•	•		0.04
•		•	•	•	•	•	0.03

Example: Posterior Effect Probability

We can compute the **model-averaged posterior** for $\mu = \mu(\beta)$:

$$\hat{f}(\mu | y) = \sum_k f(\mu | m_k, y) f(m_k | y)$$

Particularly, the **posterior effect probability** $P(\beta_i \neq 0 | y)$ is estimated by the proportion of models in which predictor i is included.

Index	Predictor	$P(\beta_i \neq 0 y)$
13	FA	1.00
14	Arb	1.00
8	RBI's	0.97
10	SOs	0.78
3	Runs	0.55
4	Hits	0.52
25	SBs $\times$ OBP	0.13
24	SOs $\times$ errors	0.12
9	Walks	0.11